

Supplementary Materials

Ecological discrepancy under a specified record-level model: 3 993 464 Mexican COVID-19 case records in 32 treating-unit states

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This document is generated from the analysis result files by `paper/make_supplement.py`. Every number below is read from those files at build time rather than transcribed, so a rerun of an analysis and a rebuild of this document cannot disagree.

Supplement A Registration and Provenance

Two protocols were frozen before the analyses they govern. The first, `ANALYSIS_PROTOCOL.md`, was frozen at commit `97f7094` and specifies ten supplementary analyses, each with a falsification criterion and a main-text or supplement placement fixed in advance. The second, `ANALYSIS_PROTOCOL_2-SITE_DEFINITIONS.md`, was frozen at commit `a38196b` and tagged `registration-s11-s12`; it governs the site definitions and the covariate ladder, records the state-level discrepancy already known at that point, and states that no grouping other than the treating unit's state had been run.

Two further records accompany them. `IMPLEMENTATION_AMENDMENT_2026-09-29.md` records changes to the implementation that do not alter the registered design, and `PROTOCOL_CORRECTION_ADDENDUM.md` records a correction to the protocol itself. Both are in the repository and both are reachable from the ledger.

Table 1: Registered analyses and the commit that froze each.

Analysis	Frozen at	Outcome known when specified
Ten supplementary analyses (S1–S10)	<code>97f7094</code>	tagged per analysis
Site definitions (S11)	<code>a38196b</code>	state-level discrepancy only
Covariate ladder (S12)	<code>a38196b</code>	state-level discrepancy only

The analysis runs recorded in the ledger carry the snapshot digest 25ccc890d190bf66..., and each names the commit it ran at and whether the analysis tree was clean.

Supplement B Denominators

The published snapshot holds 12 425 179 rows. Confirmed cases are those with CLASIFICACION_FINAL in $\{1, 2, 3\}$: $251\,234 + 14\,870 + 3\,727\,360 = 3\,993\,464$, of which 299 581 record a death. Every state-level analysis uses that denominator. The municipality analyses use fewer records because a size rule applies; 2 874 confirmed cases carry no municipality code in the catalogue.

Supplement C Registered Supplementary Analyses

Three of the ten registered analyses use the Mexico records reported in the main text. The remaining seven were specified on the two further record sets described in Supplement E and are reported there with the data they used.

S2: Mundlak within/between decomposition. A logistic model with state fixed effects and within-state and between-state elderly covariates over 3 993 464 records gives a within-state coefficient of +2.4226 (OR = 11.2746). The between-state coefficient is not identified: the state means lie in the span of the state dummies. *Falsification:* the criterion fires, and the collinearity is a property of the specification rather than evidence about aggregation.

S3: Expected ecological coefficient. Averaging the record-level model’s predicted probabilities within each state and regressing those averages on elderly share gives a model-implied slope of +0.4537 against an observed slope of +1.3105, a difference of +0.8568 and a ratio of 2.8883. The state-bootstrap interval on the difference is [+0.4778, +1.2851] over 2000 draws resampled by state. The registered agreement rule—observed and expected within 20%—does not hold.

S9: Two-stage against one-stage. Three estimators on the same records: ecological regression ($\beta = +1.3105$, $R^2 = 0.5041$); two-stage individual participant data pooled by DerSimonian–Laird (OR = 11.7293, $I^2 = 0.9741$); and the one-stage random-intercept fit reported in Supplement D.

The ecological estimator is on the mortality-rate scale and the other two on the log odds scale, so the three are not directly comparable in magnitude.

Supplement D Per-State Estimates, Weighting and Site Definitions

The covariate ladder. Each row fits the named covariates at the record level, averages the predicted probabilities within each state, and regresses those averages on elderly share.

Table 2: Model-implied slope and discrepancy by record-level specification.

Specification	Terms	Implied slope	Discrepancy
Elderly only	2	+0.3319	+0.9787
Elderly, sex, any comorbidity	4	+0.4537	+0.8568
Elderly, sex, nine comorbidities	12	+0.4949	+0.8156
Age bands, sex, nine comorbidities	22	+0.6498	+0.6610
Age spline, sex, nine comorbidities	15	+0.6549	+0.6557
Age spline, sex, nine comorbidities, month	38	+0.7164	+0.5941

Weighting and the response scale. The primary fit weights each state equally. Two alternatives are reported beside it: a fit weighting each state by its record count, and a grouped-binomial fit on the log odds scale. The three answer different questions and are not on a common scale; the log odds column is not comparable in magnitude with the two rate-scale columns.

Table 3: Discrepancy by site definition, under equal-state weighting, record-count weighting, and a grouped-binomial fit on the log odds scale.

Site definition	Sites	Observed	Implied	Equal-state	Record-count	Log odds
Treating unit's state	32	+1.3105	+0.4537	+0.8568	+1.3215	+19.302
State of residence	32	+1.3047	+0.4550	+0.8497	+1.3124	+19.268
Health-care sector	11	+1.1935	+0.4278	+0.7657	+0.9617	+8.335

Municipality at every registered threshold. On the 3 665 523 records retained by the $\geq 1,000$ rule, the paired contrast between 385 municipalities and 32 states of residence is -0.5398 .

Table 4: Municipality grouping at the four registered size thresholds.

Size rule	Sites	Records	Observed	Implied	Discrepancy
$\geq 1,000$ records	385	3 665 523	+0.8928	+0.4239	+0.4689
≥ 500 records	562	3 792 538	+0.8352	+0.4219	+0.4132
≥ 250 records	825	3 887 378	+0.7956	+0.4247	+0.3709
No size rule	2429	3 990 590	+0.4795	+0.3562	+0.1233

55 **Constrained random partitions.** Reassigning records to pseudo-sites that hold each state’s size
56 and count aged 70 or older fixed, without using any other record field, leaves a median discrepancy of
57 +0.0005 over 500 replications, with a randomization interval of $[-0.0233, +0.0224]$ and no deviation
58 from either quota. The expectation of that statistic is zero under this scheme, because the record-
59 level model is fitted by maximum likelihood and its residuals sum to zero within the strata the
60 quotas hold fixed, so the comparison is a check on the construction rather than evidence about
61 geography. Matching size alone, without the elderly quota, gives a median of -0.0152 with a wider
62 interval of $[-0.4312, +0.4096]$.

Table 5: Per-state records, elderly share, observed and model-implied mortality, and their difference, under the registered three-covariate model.

State	Records	Elderly share	Observed	Implied	Difference
1	39 537	0.0695	0.0788	0.0783	+0.0006
2	91 278	0.0749	0.1199	0.0860	+0.0340
3	63 163	0.0370	0.0381	0.0632	-0.0251
4	25 030	0.0700	0.0826	0.0812	+0.0014
5	102 613	0.0708	0.0763	0.0718	+0.0045
6	34 054	0.0618	0.0655	0.0756	-0.0102
7	22 307	0.0790	0.0957	0.0823	+0.0134
8	81 063	0.0829	0.1080	0.0849	+0.0232
9	1 154 276	0.0585	0.0458	0.0693	-0.0235
10	50 686	0.0626	0.0610	0.0692	-0.0081
11	204 282	0.0682	0.0653	0.0748	-0.0095
12	76 480	0.0822	0.0801	0.0802	-0.0002
13	61 663	0.0782	0.1262	0.0860	+0.0402
14	166 123	0.0891	0.1062	0.0885	+0.0177

State	Records	Elderly share	Observed	Implied	Difference
15	245 313	0.0722	0.1349	0.0752	+0.0597
16	73 196	0.0895	0.1101	0.0871	+0.0230
17	49 968	0.0924	0.0984	0.0838	+0.0146
18	34 670	0.0853	0.0871	0.0825	+0.0047
19	211 101	0.0629	0.0656	0.0724	-0.0068
20	83 247	0.0744	0.0651	0.0740	-0.0088
21	129 232	0.0790	0.1219	0.0807	+0.0412
22	101 452	0.0502	0.0580	0.0690	-0.0111
23	62 324	0.0359	0.0646	0.0623	+0.0023
24	110 068	0.0727	0.0619	0.0776	-0.0158
25	75 629	0.0932	0.1196	0.0872	+0.0324
26	124 098	0.0755	0.0746	0.0801	-0.0055
27	146 002	0.0500	0.0406	0.0677	-0.0271
28	105 769	0.0645	0.0697	0.0753	-0.0057
29	27 143	0.0725	0.1036	0.0756	+0.0281
30	121 775	0.0846	0.1216	0.0840	+0.0376
31	76 590	0.0694	0.0766	0.0734	+0.0032
32	43 332	0.0840	0.0782	0.0833	-0.0051

Supplement E State-Level Model Diagnostics

The state-level model is a logistic random-intercept fit over 152 919 cells with the registered covariates, the centered age spline, and the state’s elderly share entered as a contextual term. It was fitted by `lme4::glmer` at 15 adaptive Gauss–Hermite quadrature nodes in 2.63 hours. The elderly share carries an odds ratio of 1.1817 per standard deviation, with a profile interval of [1.0719, 1.3027] and a Wald interval of [1.0750, 1.2989]. The random-intercept standard deviation is 0.2682.

A second implementation written separately, using adaptive Gauss–Hermite quadrature in Python, reproduces the fit: the largest absolute coefficient difference is 7.65×10^{-5} against a tolerance of 1×10^{-4} , and the two random-intercept standard deviations are 0.2682 and 0.2682. The relative gradient at the optimum is 3.38×10^{-7} .

The result is reported only if eleven acceptance criteria hold. All eleven hold.

An earlier validation run checked the same Python implementation against `lme4` on a reduced design: the log odds agree to 8.9×10^{-8} and the standard errors to 4.14×10^{-8} , against a tolerance

Table 6: Acceptance criteria for the state-level model. A result failing any one of these is reported as not evaluable rather than as an estimate.

Criterion	Outcome
Covariates are the registered ones	pass
The optimizer reports convergence	pass
Term order matches the registered design	pass
Variance component is off the boundary	pass
The estimate is finite	pass
A second implementation reproduces it	pass
The Hessian is positive definite	pass
The Hessian agrees across step sizes	pass
The Hessian is well conditioned	pass
The gradient vanishes at the optimum	pass
A profile interval is available	pass

76 of 1×10^{-4} .

77 **Supplement F Two Further Record Sets**

78 Seven of the ten registered supplementary analyses were specified on two record sets that are not
79 part of the analysis reported in the main text: United States case surveillance published by the
80 Centers for Disease Control and Prevention (`data.cdc.gov`, dataset `vbim-akqf`), grouped by race
81 and ethnicity into pseudo-sites, and the 4CE consortium’s aggregate neurological data. They are
82 reported here because they were registered, and the main text draws no conclusion from them.
83 Their results are in `paper/analyses/results/` under `s1`, `s4` through `s8`, and `s10`.